

Tax Deferral Burden and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Tax Deferral Burden (TDB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TDB achieves an annualized gross (net) Sharpe ratio of 0.55 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 48 (46) bps/month with a t-statistic of 4.65 (4.48), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price) is 29 bps/month with a t-statistic of 2.90.

1 Introduction

Asset prices reflect firms' production decisions and the frictions they face when adjusting capital in response to economic shocks. The cross-section of stock returns reveals systematic patterns related to firms' operational characteristics, investment behavior, and exposure to aggregate productivity risks. We examine how tax deferral burden (TDB) captures variation in firms' marginal costs of production and investment frictions, thereby predicting cross-sectional differences in expected returns. Tax deferrals create wedges between accounting profits and economic cash flows, affecting firms' cost of capital and their optimal investment responses to productivity shocks. These deferrals represent accumulated timing differences that must eventually reverse, creating predictable variation in firms' future tax obligations and cash flows available for investment.

In a production-based asset pricing framework, firms with higher tax deferral burdens face elevated marginal costs of adjusting capital stock in response to productivity shocks [Cochrane and Piazzesi \(2005\)](#). Tax deferrals arise when firms accelerate depreciation for tax purposes while using straight-line depreciation for financial reporting, creating deferred tax liabilities that increase the effective cost of capital investment [Zhang \(2005\)](#). These timing differences compound investment adjustment costs, as firms must allocate future cash flows to satisfy deferred tax obligations rather than funding new productive investments. The accumulation of deferred taxes effectively increases operating leverage by creating fixed claims on future cash flows, amplifying firms' exposure to aggregate productivity shocks [Carlson et al. \(2004\)](#). When productivity declines, firms with substantial tax deferral burdens experience sharper contractions in investment opportunities because the fixed tax obligations consume a larger fraction of diminished cash flows. This mechanism generates time-varying risk premia as investors demand higher returns to compensate for bearing systematic risk associated with productivity-driven investment frictions

[Liu et al. \(2009\)](#). The tax deferral burden thus proxies for firms’ conditional exposure to investment-specific technology shocks that drive aggregate fluctuations in the marginal product of capital [Papanikolaou and Kogan \(2011\)](#).

We find that a value-weighted long/short trading strategy based on TDB achieves an annualized gross Sharpe ratio of 0.55 and delivers monthly average abnormal gross returns of 48 basis points relative to the Fama-French five-factor model plus momentum, with a t-statistic of 4.65. The strategy maintains economic significance after accounting for transaction costs, generating net abnormal returns of 46 basis points per month with a t-statistic of 4.48. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the TDB strategy earns 39 basis points per month with a t-statistic of 3.05, demonstrating that the effect persists in the most liquid segment of the market where implementation is most feasible. The strategy’s performance remains robust across alternative portfolio construction methodologies, with twenty out of twenty-five specifications producing statistically significant net alphas after accounting for trading costs. Controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the TDB strategy retains an alpha of 29 basis points per month with a t-statistic of 2.90, confirming that it captures unique variation in expected returns. The TDB signal’s gross Sharpe ratio exceeds 93% of documented anomalies in the factor zoo, while its net Sharpe ratio after transaction costs surpasses 99% of existing strategies.

Our findings contribute to the production-based asset pricing literature by identifying tax deferral burden as a novel proxy for investment frictions that generate cross-sectional return predictability. Unlike [Fama and French \(2015\)](#), who focus on profitability and investment as direct measures of firm characteristics, we demonstrate that tax-induced distortions to the cost of capital create systematic variation in risk premia through their effect on firms’ production decisions. While [Hou et al. \(2015\)](#) develop a q-factor model based on investment and profitability, our tax defer-

ral measure captures frictions in the adjustment cost function that their factors do not fully span, as evidenced by significant alphas relative to their model. Our work extends [Balakrishnan et al. \(2010\)](#) by showing that tax deferrals predict returns not through information uncertainty but through their role as impediments to optimal capital allocation in production economies. The methodological innovation lies in mapping an accounting construct—deferred tax positions—to the primitive friction of adjustment costs in production-based models, providing a bridge between the accounting literature on tax planning and the finance literature on investment-based pricing [Thomas and Zhang \(2002\)](#). The economic magnitude and statistical robustness of the TDB effect across size groups and after controlling for transaction costs establishes its relevance for practical portfolio construction. These results deepen our understanding of how tax policy and accounting rules interact with firms’ real investment decisions to generate predictable variation in the cross-section of expected returns, with implications for both asset pricing theory and the evaluation of tax policy effects on capital markets.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over several decades, ensuring broad cross-sectional and time-series coverage of the U.S. equity market. We focus on annual observations from fiscal year-end financial statements to construct our Tax Deferral Burden signal. Tax Deferral Burden signal requires two key accounting variables. Deferred taxes federal (TXDFED) represents the federal portion of deferred income tax expense reported in a firm’s income statement, capturing the difference between tax expense recognized for financial reporting

purposes and taxes actually paid to federal authorities during the fiscal year. This item reflects timing differences between book and tax accounting that will reverse in future periods. Earnings before interest and taxes (EBIT) measures a firm’s operating performance before the effects of capital structure and tax considerations, calculated as revenues minus operating expenses including depreciation and amortization but excluding interest expenses and income taxes. We construct the Tax Deferral Burden signal by dividing deferred taxes federal (TXDFED) by earnings before interest and taxes (EBIT). This ratio captures the proportion of a firm’s operating earnings that represents deferred federal tax obligations, providing a scaled measure of the tax timing differences relative to the firm’s operating performance. We calculate this signal annually using fiscal year-end values and require non-missing values for both TXDFED and EBIT. To ensure meaningful ratios, we exclude observations where EBIT equals zero or is negative, as these cases would produce undefined or economically uninterpretable values. The resulting signal measures the extent to which a firm’s current tax expense differs from its cash tax payments, scaled by its pre-tax operating profitability.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TDB signal. Panel A plots the time-series of the mean, median, and interquartile range for TDB. On average, the cross-sectional mean (median) TDB is 0.00 (-0.00) over the 1985 to 2024 sample, where the starting date is determined by the availability of the input TDB data. The signal’s interquartile range spans -0.09 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the TDB signal for the CRSP universe. On average, the TDB signal is available for 5.37% of CRSP names, which on average make up 5.90% of total market capitalization.

4 Does TDB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TDB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TDB portfolio and sells the low TDB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TDB strategy earns an average return of 0.37% per month with a t-statistic of 3.46. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.34% to 0.48% per month and have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.27, with a t-statistic of -5.92 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 391 stocks and an average market capitalization of at least \$1,389 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.19. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -5-34bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TDB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TDB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TDB, as well as average returns and alphas for long/short trading TDB strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TDB strategy achieves an average return of 39 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the TDB strategy relative to the five most common factor models range from 33 to 52 bps/month with t-statistics between 2.54 and 4.32.

5 How does TDB perform relative to the zoo?

Figure 2 puts the performance of TDB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TDB strategy falls in the distribution. The TDB strategy’s gross (net) Sharpe ratio of 0.55 (0.50) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TDB strategy (red line).² Ignoring trading costs, a \$1 invested in the TDB strategy would have yielded \$4.24 which ranks the TDB strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TDB strategy would have yielded \$3.55 which ranks the TDB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the TDB relative to those. Panel A shows that the TDB strategy gross alphas fall between the 66 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198506 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TDB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TDB ranks between the 83 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TDB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of TDB with 209 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TDB or at least to weaken the power TDB has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of TDB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDB}TDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TDB. Stocks are finally grouped into five TDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TDB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TDB signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for Equity Duration. Controlling for all six closely related anomalies, the t-statistic on TDB is 0.77.

Similarly, Table 5 reports results from spanning tests that regress returns to the TDB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TDB strategy earns alphas that range from 28-51bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.84,

which is achieved when controlling for Equity Duration. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TDB trading strategy achieves an alpha of 29bps/month with a t-statistic of 2.90.

7 Does TDB add relative to the whole zoo?

Finally, we can ask how much adding TDB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the TDB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$46.04, while \$1 investment in the combination strategy that includes TDB grows to \$60.34.

8 Conclusion

This study provides compelling evidence that Tax Deferral Burden (TDB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TDB-based trading strategies generate substantial

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TDB is available.

risk-adjusted returns that persist even after controlling for established risk factors and related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.55 gross (0.50 net), with monthly alphas of 48 bps relative to the Fama-French five-factor model plus momentum, maintaining statistical significance with t-statistics exceeding 4.48. Particularly noteworthy is the strategy’s resilience when tested against a comprehensive set of twelve factors, including six closely related strategies from the factor zoo, where it still delivers a statistically significant alpha of 29 bps per month (t-statistic = 2.90). These findings suggest that TDB captures a distinct source of mispricing not fully explained by existing factors, potentially reflecting market inefficiencies in processing the implications of deferred tax positions on firm value and future cash flows. From a practical perspective, the modest deterioration from gross to net returns indicates that implementation costs do not substantially erode the strategy’s profitability, enhancing its appeal for institutional investors. However, several limitations warrant consideration. First, the analysis is confined to U.S. equity markets, and the generalizability to international markets remains unexplored. Second, the stability of TDB’s predictive power across different market regimes and economic cycles requires further investigation. Third, while transaction costs are incorporated, other practical considerations such as market impact and capacity constraints deserve deeper examination. Future research should explore the economic mechanisms underlying the TDB anomaly, investigate its interaction with corporate tax policy changes, and examine whether similar tax-related inefficiencies exist in other asset classes. Additionally, understanding the decay of the signal’s effectiveness over time and its relationship with investor sophistication and institutional ownership could provide valuable insights into the persistence of this anomaly.

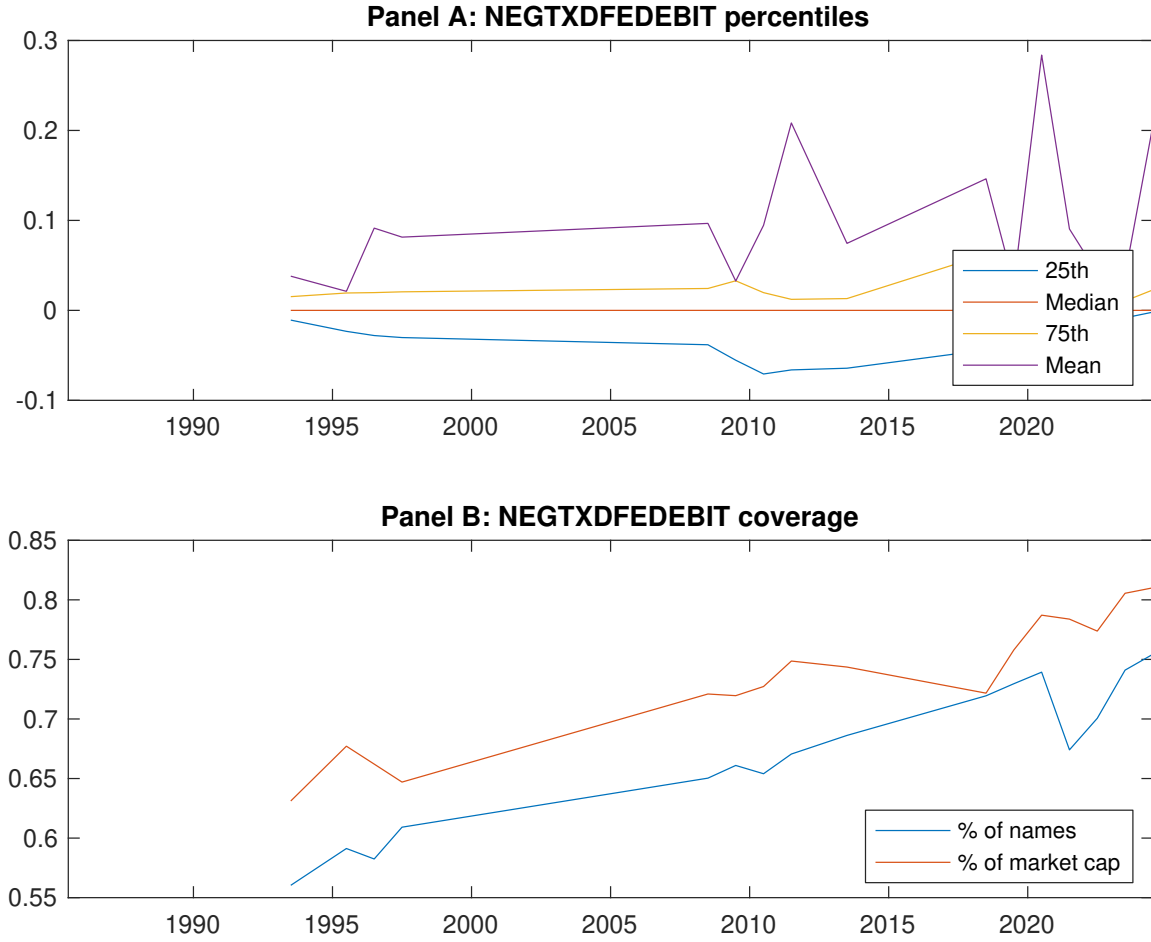


Figure 1: Times series of TDB percentiles and coverage. This figure plots descriptive statistics for TDB. Panel A shows cross-sectional percentiles of TDB over the sample. Panel B plots the monthly coverage of TDB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TDB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Excess returns and alphas on TDB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.37]	0.82 [3.95]	0.70 [3.21]	0.73 [3.07]	0.93 [3.81]	0.37 [3.46]
α_{CAPM}	-0.25 [-3.14]	0.11 [1.61]	-0.05 [-0.62]	-0.09 [-1.42]	0.10 [1.18]	0.34 [3.18]
α_{FF3}	-0.27 [-3.57]	0.09 [1.32]	-0.02 [-0.26]	-0.07 [-1.04]	0.14 [1.79]	0.41 [4.05]
α_{FF4}	-0.23 [-3.10]	0.13 [1.88]	-0.00 [-0.06]	-0.07 [-1.04]	0.17 [2.18]	0.41 [3.98]
α_{FF5}	-0.27 [-3.49]	-0.02 [-0.33]	-0.07 [-1.00]	-0.04 [-0.56]	0.21 [2.72]	0.48 [4.71]
α_{FF6}	-0.24 [-3.13]	0.01 [0.21]	-0.06 [-0.82]	-0.04 [-0.57]	0.24 [3.06]	0.48 [4.65]
Panel B: Fama and French (2018) 6-factor model loadings for TDB-sorted portfolios						
β_{MKT}	1.07 [58.44]	0.97 [61.65]	0.98 [57.10]	1.09 [69.41]	1.07 [58.00]	-0.00 [-0.11]
β_{SMB}	0.11 [3.87]	-0.01 [-0.39]	0.15 [5.66]	-0.06 [-2.42]	0.02 [0.61]	-0.09 [-2.44]
β_{HML}	0.08 [2.29]	-0.05 [-1.74]	-0.13 [-4.00]	-0.13 [-4.53]	-0.19 [-5.58]	-0.27 [-5.92]
β_{RMW}	-0.03 [-0.86]	0.23 [7.71]	0.15 [4.57]	-0.10 [-3.27]	-0.25 [-7.00]	-0.22 [-4.62]
β_{CMA}	0.10 [2.09]	0.11 [2.55]	-0.02 [-0.38]	0.04 [0.99]	0.17 [3.44]	0.07 [1.03]
β_{UMD}	-0.05 [-2.98]	-0.07 [-4.39]	-0.02 [-1.41]	0.00 [0.13]	-0.05 [-2.73]	0.00 [0.17]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	391	910	703	508	
me (\$10 ⁶)	1389	2381	2463	2846	2480	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TDB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37 [3.46]	0.34 [3.18]	0.41 [4.05]	0.41 [3.98]	0.48 [4.71]	0.48 [4.65]
Quintile	NYSE	EW	0.13 [2.19]	0.10 [1.72]	0.12 [2.09]	0.11 [1.94]	0.11 [1.90]	0.11 [1.80]
Quintile	Name	VW	0.26 [2.44]	0.21 [1.98]	0.27 [2.80]	0.27 [2.70]	0.37 [3.75]	0.36 [3.64]
Quintile	Cap	VW	0.34 [2.98]	0.28 [2.47]	0.35 [3.36]	0.34 [3.23]	0.47 [4.59]	0.46 [4.45]
Decile	NYSE	VW	0.38 [2.58]	0.38 [2.57]	0.45 [3.15]	0.40 [2.79]	0.53 [3.66]	0.49 [3.37]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.34 [3.17]	0.32 [2.98]	0.37 [3.72]	0.37 [3.69]	0.46 [4.55]	0.46 [4.48]
Quintile	NYSE	EW	-0.05 [-0.67]					
Quintile	Name	VW	0.23 [2.18]	0.19 [1.80]	0.24 [2.49]	0.24 [2.44]	0.35 [3.56]	0.34 [3.50]
Quintile	Cap	VW	0.31 [2.74]	0.25 [2.17]	0.31 [2.91]	0.30 [2.85]	0.43 [4.24]	0.43 [4.14]
Decile	NYSE	VW	0.34 [2.31]	0.35 [2.33]	0.40 [2.82]	0.37 [2.63]	0.50 [3.44]	0.47 [3.28]

Table 3: Conditional sort on size and TDB

This table presents results for conditional double sorts on size and TDB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TDB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TDB and short stocks with low TDB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198506 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TDB Quintiles					TDB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.82	0.55	0.20	0.55	0.98	0.16	0.15	0.17	0.14	0.08	0.06
		[2.81]	[1.75]	[0.49]	[1.50]	[3.13]	[1.30]	[1.19]	[1.31]	[1.05]	[0.62]	[0.49]
	(2)	0.74	0.87	0.39	0.60	0.96	0.22	0.19	0.22	0.21	0.25	0.24
		[2.64]	[3.20]	[1.04]	[1.99]	[3.23]	[2.27]	[1.95]	[2.39]	[2.26]	[2.61]	[2.51]
	(3)	0.76	0.78	0.65	0.81	0.84	0.08	0.07	0.12	0.10	0.15	0.14
	[2.70]	[3.09]	[1.95]	[2.86]	[3.00]	[0.70]	[0.56]	[1.04]	[0.92]	[1.29]	[1.18]	
(4)	0.89	0.78	0.82	0.85	0.94	0.05	0.00	0.08	0.04	0.22	0.18	
	[3.43]	[3.27]	[2.88]	[3.20]	[3.39]	[0.39]	[0.01]	[0.70]	[0.35]	[1.82]	[1.51]	
(5)	0.54	0.81	0.75	0.75	0.92	0.39	0.33	0.39	0.39	0.52	0.52	
	[2.36]	[3.98]	[3.57]	[3.30]	[3.77]	[3.05]	[2.54]	[3.20]	[3.16]	[4.32]	[4.26]	
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TDB Quintiles					TDB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	345	341	336	337	346	41	37	29	32	42	
	(2)	97	97	97	97	96	66	68	64	65	66	
	(3)	63	64	64	64	63	110	111	109	112	110	
	(4)	51	51	51	52	51	234	241	236	239	239	
(5)	48	48	48	48	48	1503	1753	1993	2008	2052		

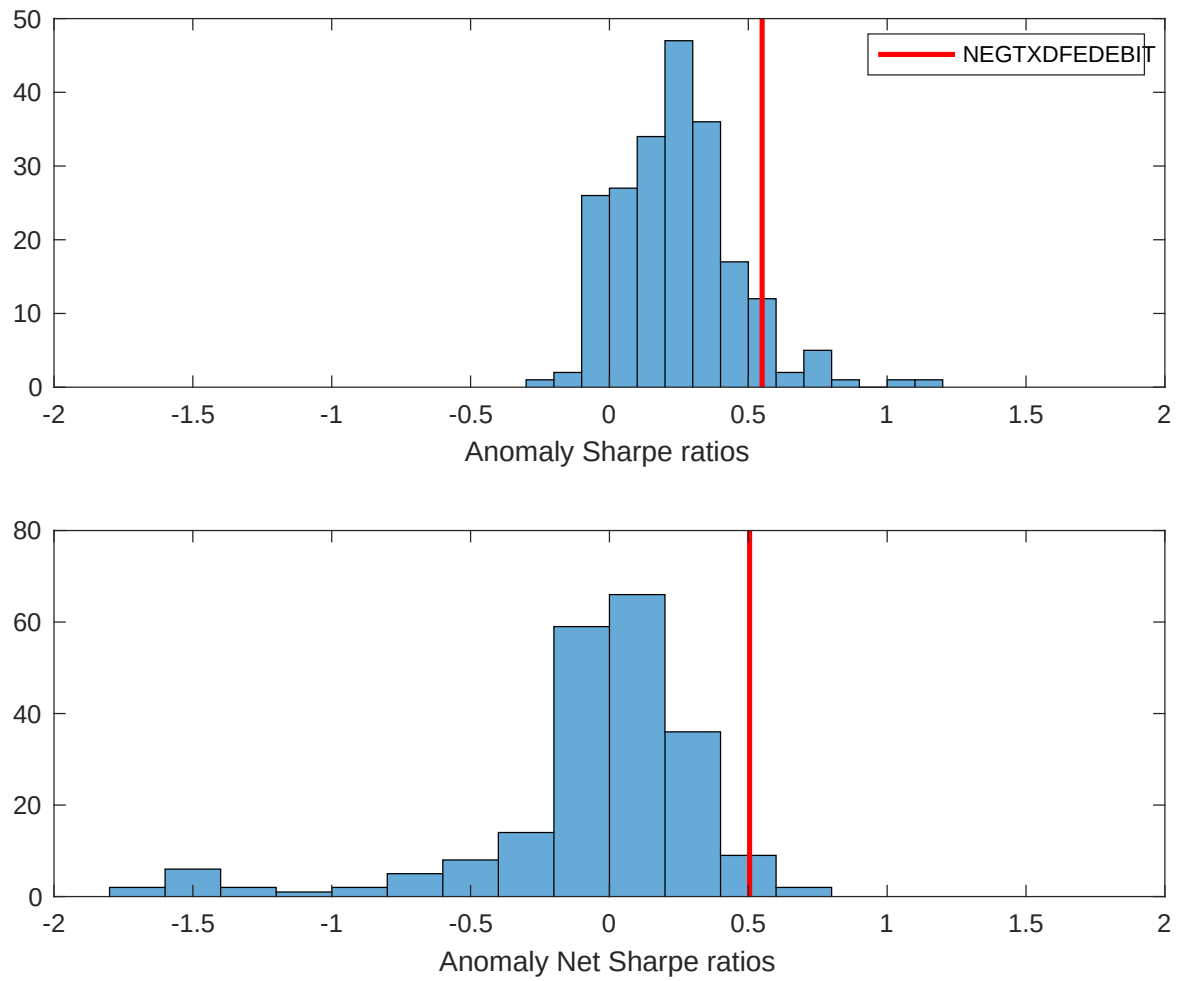


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TDB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

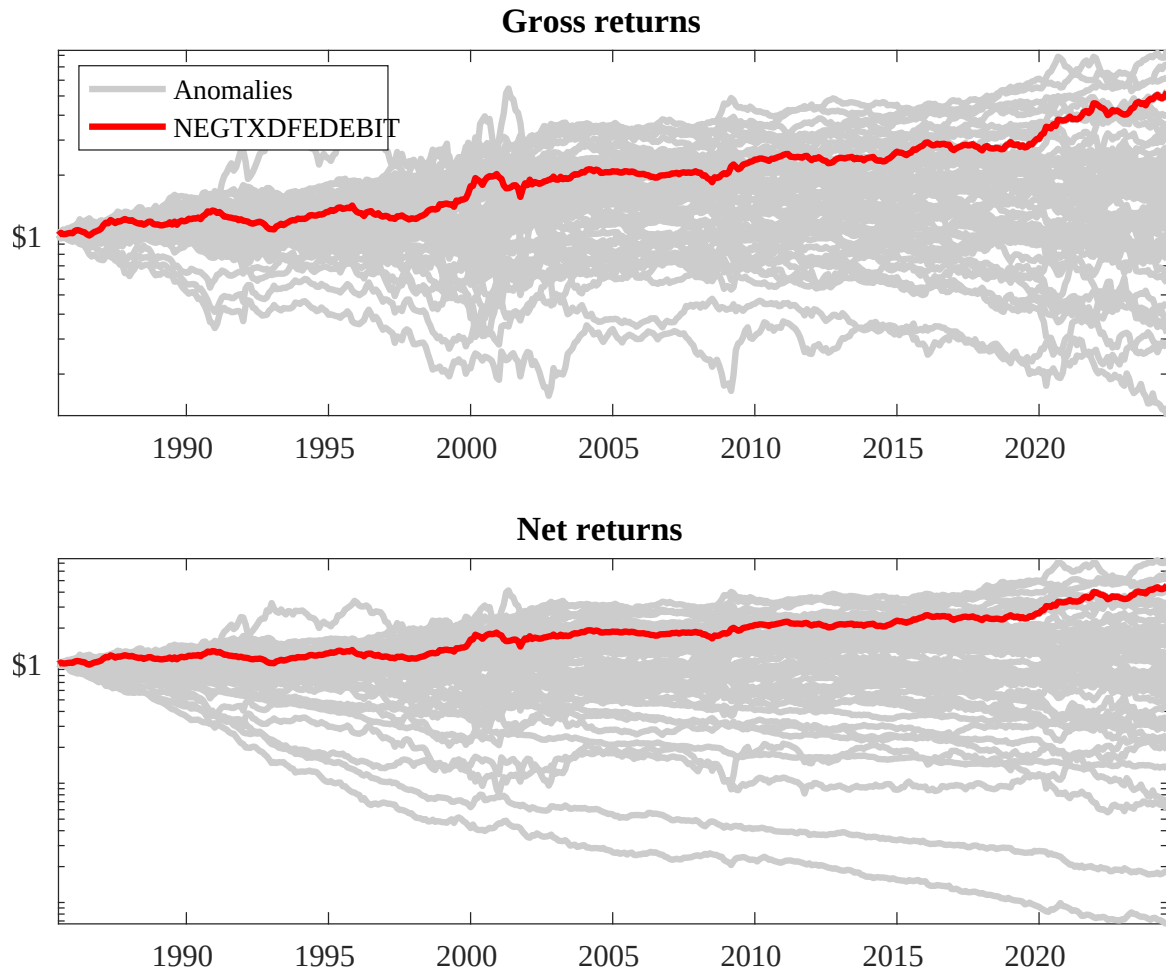
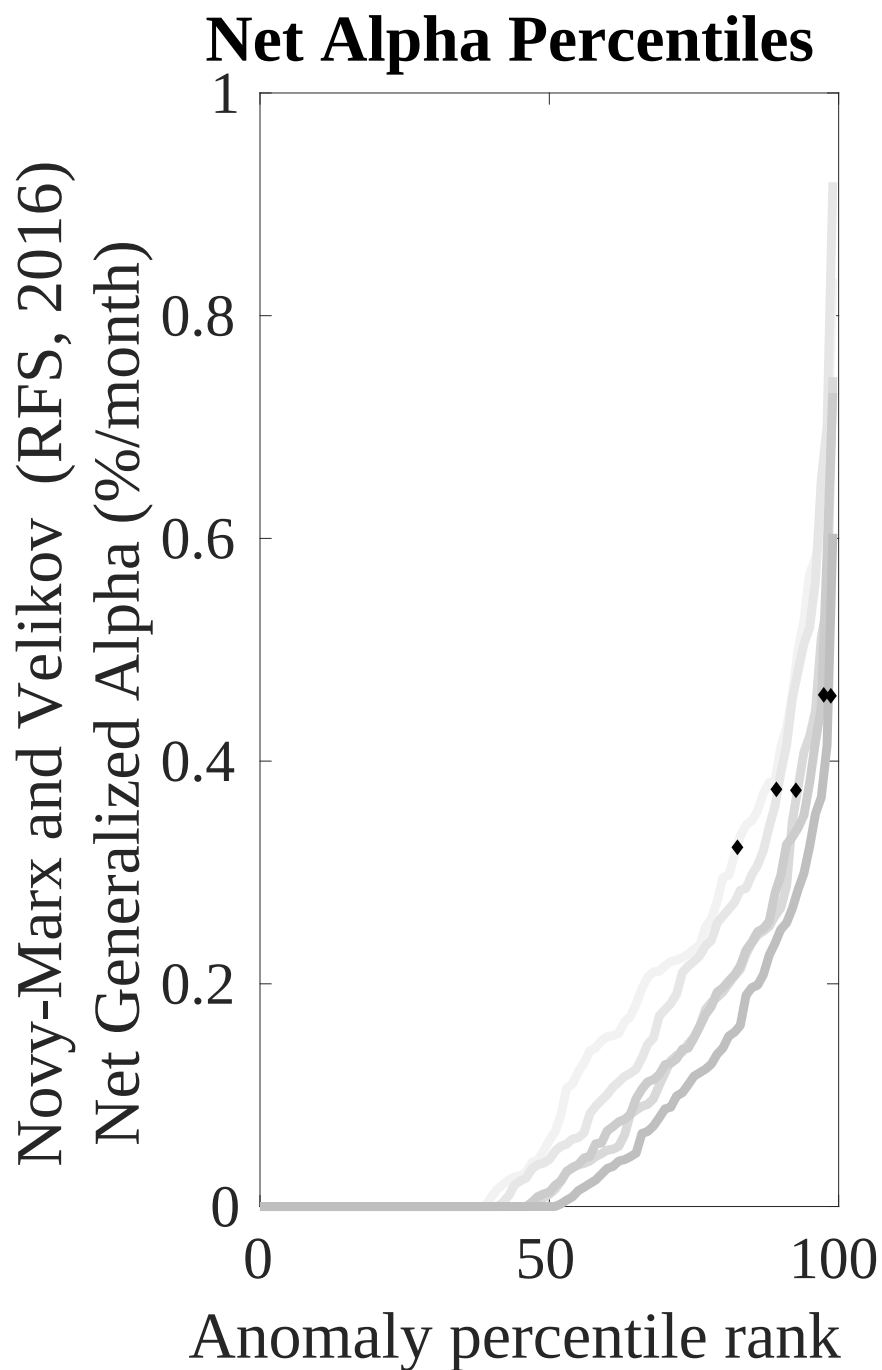
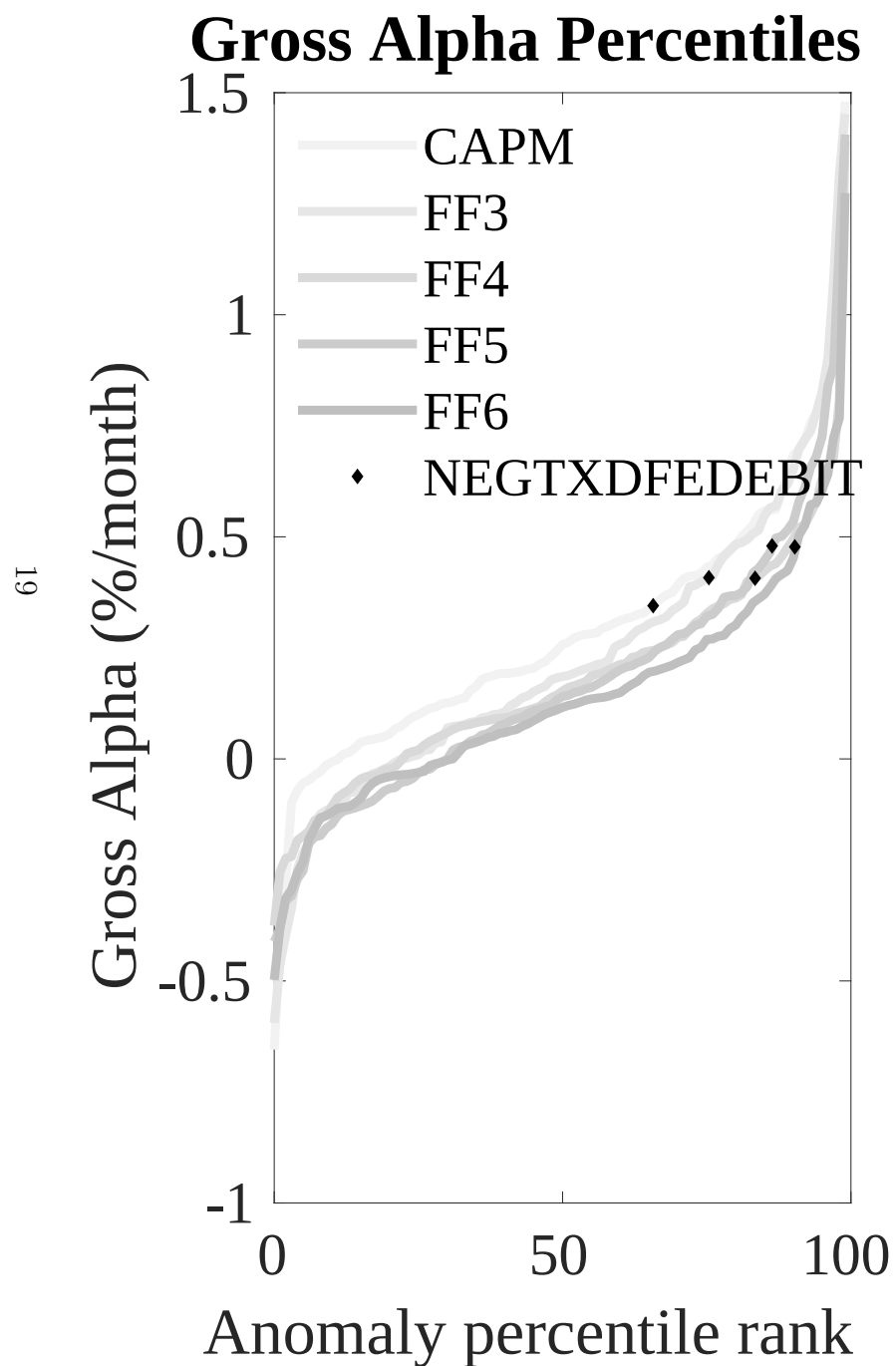


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TDB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



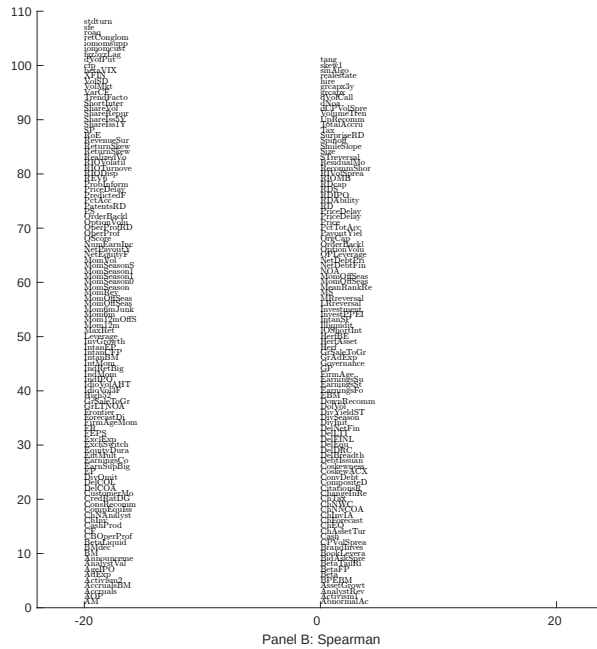
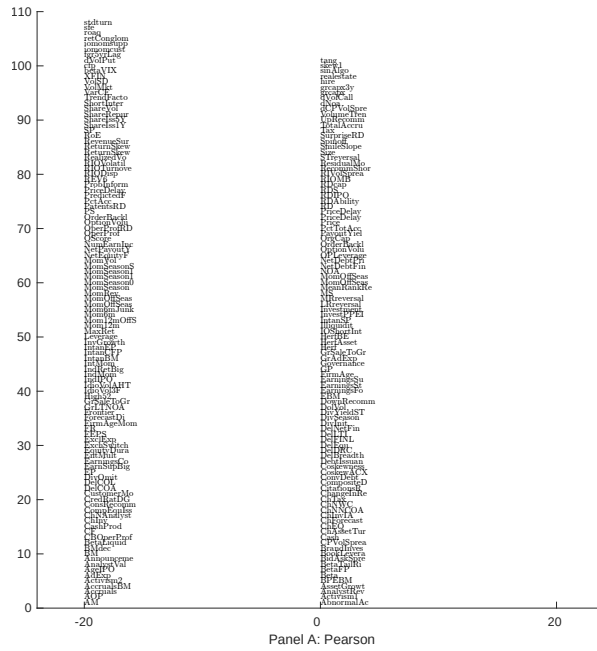


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with TDB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

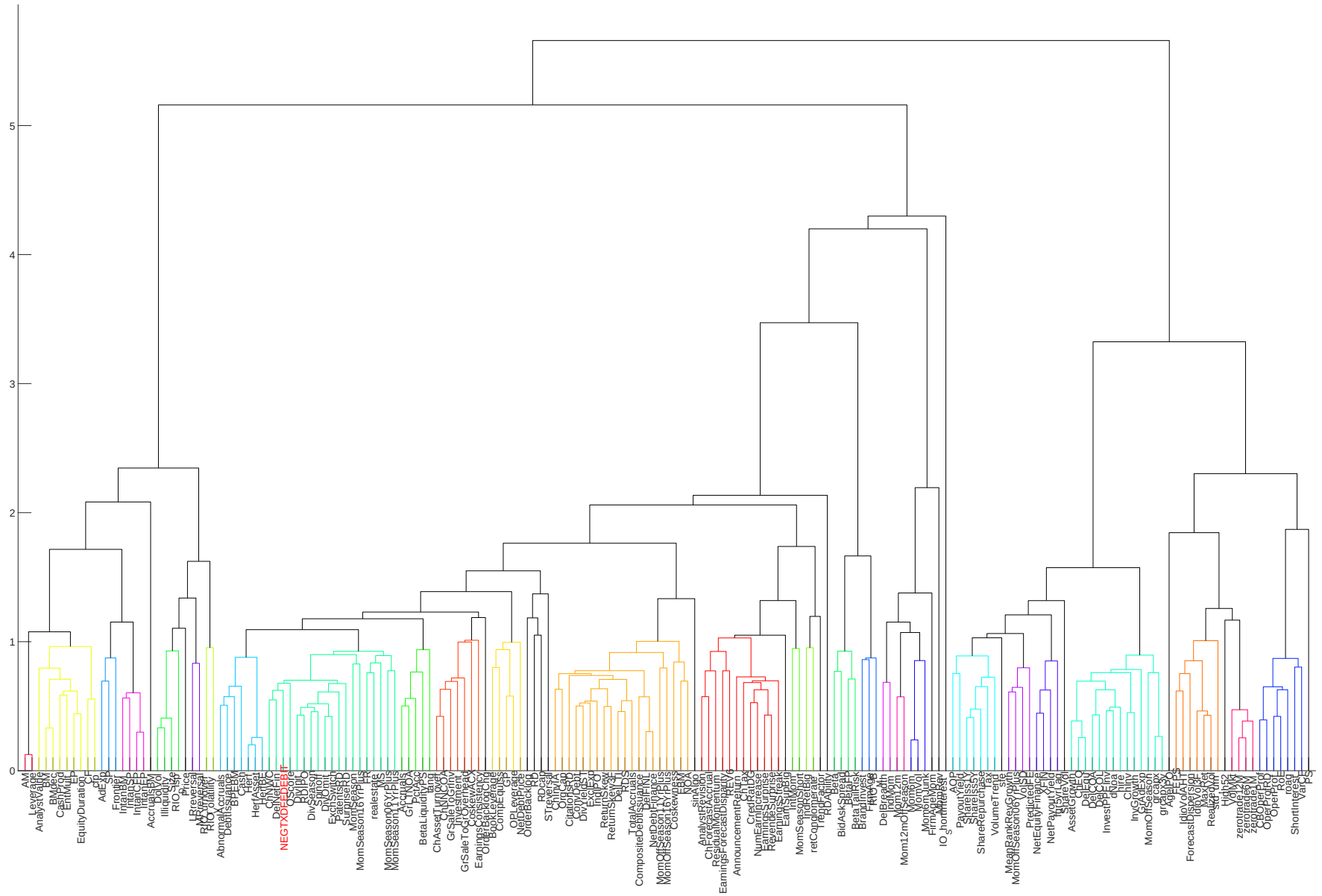


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

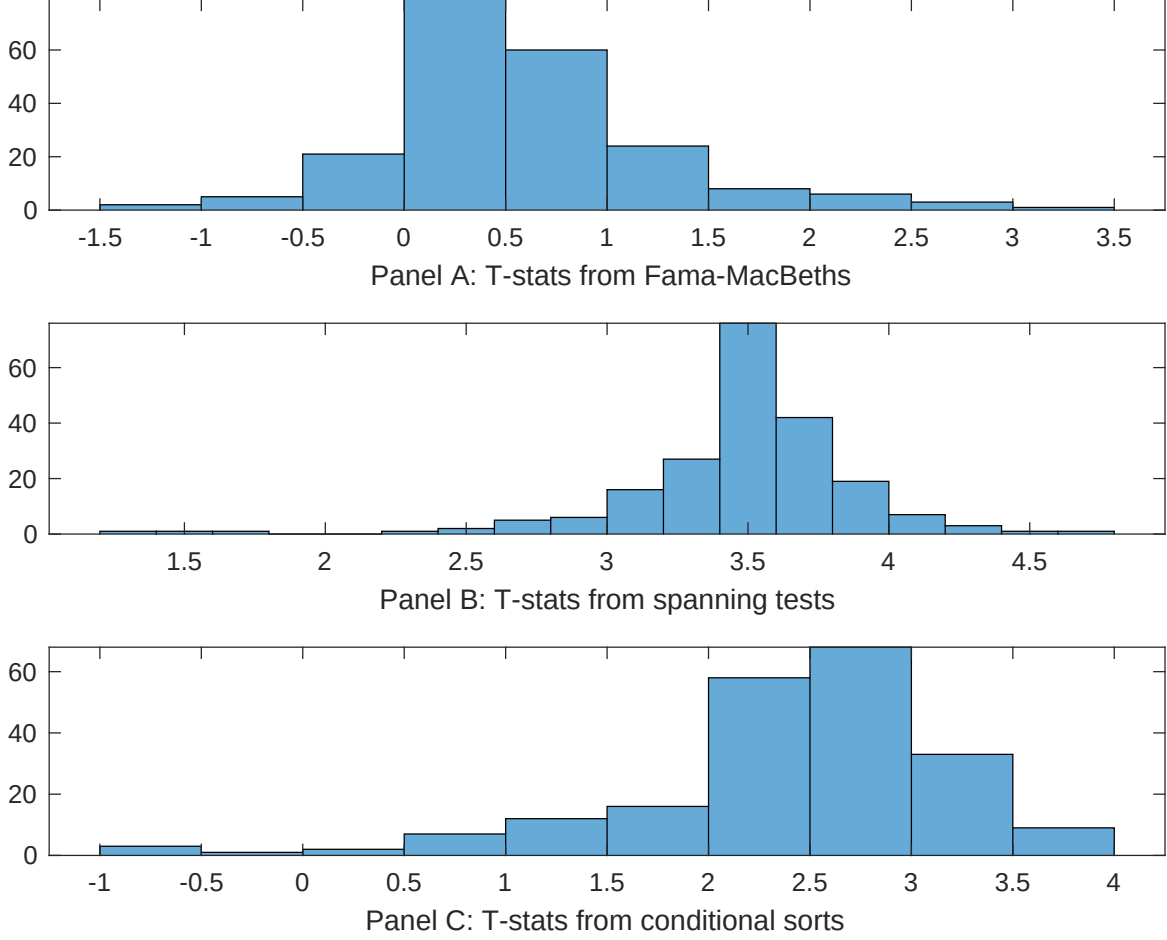


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TDB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDB}TDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TDB. Stocks are finally grouped into five TDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TDB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TDB}TDB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.10 [3.69]	0.16 [6.32]	0.10 [3.50]	0.93 [3.04]	0.10 [3.51]	0.10 [3.46]	0.12 [4.81]
TDB	0.78 [1.35]	0.18 [0.31]	0.24 [0.43]	0.34 [0.59]	0.39 [0.69]	0.34 [0.59]	0.43 [0.77]
Anomaly 1	0.62 [1.56]						0.11 [3.13]
Anomaly 2		0.25 [2.66]					0.23 [3.41]
Anomaly 3			0.18 [0.58]				-0.23 [-0.98]
Anomaly 4				0.27 [3.22]			0.20 [2.84]
Anomaly 5					0.13 [0.61]		-0.25 [-1.32]
Anomaly 6						0.85 [2.95]	0.79 [4.36]
# months	463	468	463	463	463	463	463
$\bar{R}^2(\%)$	1	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TDB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TDB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.28 [2.84]	0.47 [4.63]	0.48 [4.73]	0.46 [4.56]	0.47 [4.56]	0.51 [5.00]	0.29 [2.90]
Anomaly 1	33.90 [8.54]						30.37 [7.01]
Anomaly 2		-13.35 [-2.49]					-1.87 [-0.29]
Anomaly 3			-8.48 [-2.40]				2.14 [0.48]
Anomaly 4				-25.89 [-3.87]			-9.98 [-1.45]
Anomaly 5					-12.58 [-3.00]		-6.98 [-1.62]
Anomaly 6						-13.55 [-3.62]	-3.78 [-0.87]
mkt	-6.58 [-2.77]	-0.34 [-0.14]	-0.21 [-0.09]	-1.30 [-0.54]	1.38 [0.56]	0.27 [0.11]	-5.33 [-2.14]
smb	-6.87 [-2.00]	-5.10 [-1.33]	-7.16 [-1.95]	-0.88 [-0.22]	-7.09 [-1.94]	-5.87 [-1.60]	-3.35 [-0.86]
hml	-12.23 [-2.75]	-10.22 [-1.36]	-19.96 [-4.06]	1.74 [0.21]	-8.23 [-1.16]	-14.92 [-2.86]	9.45 [1.00]
rmw	-5.99 [-1.26]	-22.11 [-4.77]	-19.30 [-4.01]	-29.19 [-5.93]	-24.13 [-5.18]	-17.73 [-3.71]	-10.77 [-1.97]
cma	18.37 [2.92]	1.37 [0.20]	8.59 [1.29]	6.95 [1.07]	3.20 [0.49]	8.92 [1.36]	15.48 [2.28]
umd	0.42 [0.20]	0.48 [0.21]	-2.98 [-1.07]	1.58 [0.69]	-2.06 [-0.83]	-4.75 [-1.73]	-1.50 [-0.49]
# months	464	468	464	464	464	464	464
$\bar{R}^2(\%)$	29	19	18	20	19	19	29

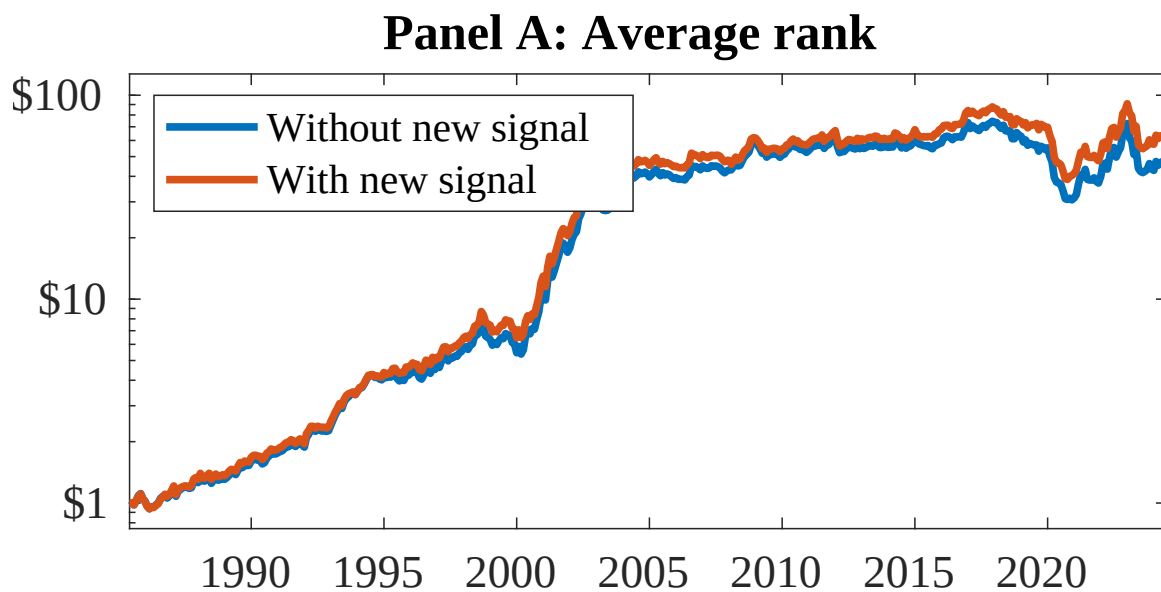


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as TDB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Balakrishnan, K., Bartov, E., and Faurel, L. (2010). Post loss/profit announcement drift. *Journal of Accounting and Economics*, 50(1):20–41.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Carlson, M., Fisher, A., and Giammarino, R. (2004). Corporate investment and asset price dynamics: Implications for the cross-section of returns. *Journal of Finance*, 59(6):2577–2603.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Papanikolaou, D. and Kogan, L. (2011). Growth opportunities and technology shocks. *American Economic Review*, 101(3):532–536.
- Thomas, J. K. and Zhang, H. (2002). Inventory changes and future returns. *Review of Accounting Studies*, 7(2):163–187.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.